



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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COMMON HALF YEARLY EXAMINATION - 2024-25

Time Allowed : 2.30 Hours

MATHEMATICS

[Max. Marks : 100]

PART - I

I. Choose the correct Answer.

10x1=10

- $-\frac{5}{4}$ is a rational number which lies between -----
a) 0 and $-\frac{5}{4}$ b) -1 and 0 c) -1 and -2 d) -4 and -5
- $\sqrt{128} - \sqrt{98} + \sqrt{18} =$ -----
a) $\sqrt{2}$ b) $\sqrt{8}$ c) $\sqrt{48}$ d) $\sqrt{32}$
- If $\frac{10^x}{10^{-3}} = 10^9$, then x is -----
a) 4 b) 5 c) 6 d) 7
- A line segment which joins any two points on a circle is a -----
a) Diameter b) Radius c) Chord d) Circular arc
- The cross section of a Solid Cylinder is -----
a) Circle b) Cube c) Cuboid d) Cylinder
- $(a-b) = 3$ and $ab = 5$ then $a^3 - b^3 =$ -----
a) 15 b) 18 c) 62 d) 72
- The largest number of the three consecutive numbers is $x+1$, then the smallest number is
a) x b) $x+1$ c) $x+2$ d) $x-1$
- If 5 persons can do 5 jobs in 5 days then 50 persons can do 50 jobs in ----- days.
a) 10 b) 5 c) 50 d) 1
- Two similar triangles will always have ----- angles.
a) Acute b) Obtuse c) right d) matching
- How many 2 digit numbers contain the number 7?
a) 10 b) 18 c) 19 d) 20

II. Fill in the Blanks.

5x1=5

- The cube root of 0.000004913 is -----
- The x - coordinate is always ----- on the y - axis.
- 2 minutes is ----- % to an hour.
- The medians of a triangle cross each other at -----
- If the compound interest is calculated Quarterly the amount is found using the formula -----

III. Say true or false.

5x1=5

- All rational numbers have an additive inverse.
- A part of circumference of a circle is called as circular arc.
- The co-ordinates of the origin are (1,1)
- Depreciation value is calculated by the formula $P \left(1 - \frac{r}{100}\right)^n$
- The centroid, ortho centre and incentre of a triangle are collinear.

IV. Match the following.

5x1=5

- | | | |
|------------------------------------|---|-------------------|
| 21. Perimeter of the sector | - | Inverse Variation |
| 22. Circumference of a semi circle | - | Direct Variation |
| 23. Purchase - spending | - | $l + 2r$ |
| 24. Price - consumption | - | 5, 12, 13 |
| 25. Pythagorean triplets | - | $(\pi + 2)r$ |

PART - II

V. Answer any 10 of the following. (Q.No.40 Compulsory)

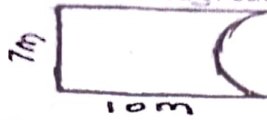
10x2=20

- Find two rational numbers between -2 and 0
- Write the decimal form of rational number $1\frac{2}{5}$
- Divide: $-\frac{21}{5}$ by $-\frac{7}{10}$
- Find the square root of 256 by prime factorisation method.
- Evaluate: $(2^{-5} \times 2^7) \div 2^{-2}$





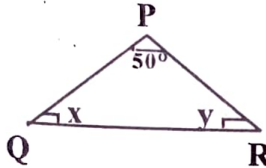
31. A spinner of radius 7.5 cm is divided into 6 equal sectors. Find the area of each of the sectors.
32. Find the area of the given Figure.



33. Using Euler's Formula, find the unknowns.

	Faces	Vertices	Edges
i)	?	6	14
ii)	8	?	10

34. Solve: $2x+5=9$
35. Factorise: $49x^2 - 64y^2$
36. Find the Difference in C.I and S.I for Principle ₹5000 $r = 4\%$, $n = 2$ years.
37. If the selling price of 10 rulers is the same as the cost price of 15 rulers, then find the profit %?
38. Find the unknown angles of the given figure.



39. If you have 2 school bags and 3 water bottles then in how many different ways can you choose each one of them, while going to school?
40. i) Find the HCF by using repeated subtraction method 42 and 70 (OR)
ii) Find the Quadrant without plotting the points on a graph sheets. (3, -4), (-3, -5), (0, 10), (5, 7)

PART - III

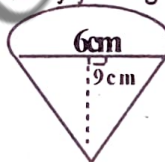
- VI. Answer any Seven questions. (Q.No.50 Compulsory.)

7x5=35

41. Write the following rational numbers in ascending and descending order:

$\frac{-5}{12}$, $\frac{-11}{8}$, $\frac{-15}{24}$, $\frac{-7}{-9}$, $\frac{12}{36}$

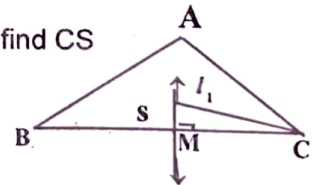
42. Find the Cube root of $24 \times 36 \times 80 \times 25$
43. The radius of sector 16 cm and its central angle is 45° . Find the length of arc, area and Perimeter.
44. Find the area of the Combined figure given, formed by joining a semicircle of diameter 6cm with a triangle of base 6cm and height = 9cm. ($\pi = 3.14$)



45. Find the volume of cuboid whose dimensions are $(x+2)$, $(x-1)$ and $(x-3)$.
46. A total of 90 currency notes, consisting only of ₹5 and ₹10 denominations, amount to ₹500. Find the number of notes in each denominations.
47. Find the total bill amount for the data given below.

S.No	Name of the item	Marked price	Discount	GST
i)	School bag	₹500	5%	12%
ii)	Hair dryer	₹2000	10%	28%

48. If 6 container lorries can transport 135 tonnes of goods in 5 days. How many more lorries are required to transport 180 tonnes of goods in 4 days?
49. In triangle ABC, line l, is a perpendicular of BC if $BC = 12\text{cm}$, $SM = 8\text{cm}$ find CS



50. i) To find the HCF 392 and 256 using repeated division method. (OR)
ii) Divide $5xy^2 - 18x^2y^3 + 6xy$ by $6xy$

PART - IV

- VII. Answer any one of the following.

1x10=10

51. a) Construct a quadrilateral DEAR with $DE = 6\text{cm}$, $EA = 5\text{cm}$, $AR = 5.5\text{cm}$, $RD = 5.2\text{cm}$, $DA = 10\text{cm}$. Also find its area.
b) Construct a Parallelogram BIRD with $BI = 6.5\text{cm}$ and $IR = 5\text{cm}$ and $\angle BIR = 70^\circ$. Also find its area.

PART - V

- VIII. Answer one of the following.

1x10=10

52. a) If the points $P(5,3)$, $Q(-3,3)$, $R(-3, -4)$ and S form a rectangle, then find the co-ordinate of S .
b) Draw the graph of $x = -7$.

(OR)

